

11. Reverse Array Without Using Another Array

Input:

1 2 3 4 5

Output:

5 4 3 2 1

```
Main.java
1 import java.util.*;
2 public class Main
3 {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6         int n = sc.nextInt();
7         int[] arr = new int[n];
8         for(int i=0;i<n;i++){
9             arr[i] = sc.nextInt();
10
11         //System.out.print(arr[i]+ " ");
12     }
13     int start = 0;
14     int end = n-1;
15     while(start < end){
16         int temp = arr[start];
17         arr[start] = arr[end];
18         arr[end] = temp;
19         start++;
20         end--;
21     }
22     for(int i=0;i<n;i++){
23
24         System.out.print(arr[i]+ " ");
25     }
26
27

```

input

5
1 2 3 4 5
5 4 3 2 1

...Program finished with exit code 0
Press ENTER to exit console.

12. Find Maximum Difference Between Two Elements

(Second element should come after first.)

Input:

2 3 10 6 4 8 1

Output:

Maximum Difference = 8

```
Main.java
1 import java.util.*;
2 public class Main
3 {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6         int n = sc.nextInt();
7         int[] arr = new int[n];
8         for(int i = 0; i < n; i++){
9             arr[i] = sc.nextInt();
10        }
11        int min = arr[0];
12        int maxdiff = arr[1] - arr[0];
13        for(int i = 1; i < n; i++){
14            if(arr[i] - arr[0] > maxdiff){
15                maxdiff = arr[i] - arr[0];
16            }
17            if (arr[i] < min) {
18                min = arr[i];
19            }
20        }
21
22        System.out.print("Maximum Difference = "+maxdiff);
23    }
24 }
25
```

input

```
7
2 3 10 6 4 8 1
Maximum Difference =8

...Program finished with exit code 0
Press ENTER to exit console.
```

13. Leaders in an Array

(A leader is greater than all elements to its right.)

Input:

16 17 4 3 5 2

Output:

17 5 2

```
Main.java
1 import java.util.*;
2 public class Main
3 {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6         int n = sc.nextInt();
7         int[] arr = new int[n];
8         for(int i=0;i<n;i++){
9             arr[i] = sc.nextInt();
10        }
11        for(int i=0;i<n;i++){
12            int count = 0;
13            for(int j=i+1;j<n;j++){
14                if(arr[i]>arr[j]){
15                    count++;
16                }
17            }
18            if(count == n-i-1){
19                System.out.print(arr[i]+" ");
20            }
21        }
22    }
23 }
24
25
```

input

```
6
16 17 4 3 5 2
17 5 2

...Program finished with exit code 0
Press ENTER to exit console.
```

14. Find Equilibrium Index

(Index where left sum = right sum.)

Input:

1 3 5 2 2

Output:

Equilibrium Index = 2

```
Main.java
1 import java.util.*;
2 public class Main
3 {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6         int n = sc.nextInt();
7         int[] arr = new int[n];
8         for(int i=0; i<n; i++){
9             arr[i] = sc.nextInt();
10        }
11        for(int i=0; i<n; i++){
12            int leftsum=0;
13            int rightsum =0;
14            for(int j=0; j<i; j++){
15                leftsum += arr[j];
16            }
17            for(int j=i+1; j<n; j++){
18                rightsum += arr[j];
19            }
20            if(leftsum == rightsum){
21                System.out.println("Equilibrium Index = "+ i);
22                return;
23            }
24        }
25        System.out.println("No Equilibrium Index");
26    }
27 }
```

input

5
1 3 5 2 2
Equilibrium Index = 2

15. Find Pair with Given Sum

Input:

Array: 2 7 11 15 5

Target: 20

Output:

Pair = 15 5

```
Main.java
1 import java.util.*;
2 public class Main
3 {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6         int n = sc.nextInt();
7         int[] arr = new int[n];
8         for(int i=0; i<n; i++){
9             arr[i] = sc.nextInt();
10        }
11        for(int i=0; i<n; i++){
12            int sum=0;
13            int Target = 9;
14            for(int j=1; j<n; j++){
15                sum = arr[i]+arr[j];
16                if(sum == Target){
17                    System.out.print("Pair =" + arr[i]+" "+arr[j]);
18                    return;
19                }
20            }
21        }
22    }
23 }
```

input

5
2 7 11 15 5
Pair =2 7

16. Find Majority Element

(Element occurring more than $n/2$ times.)

Input:

2 2 1 2 3 2 2

Output:

Majority Element = 2

```
Main.java  ⋮
1  import java.util.*;
2  public class Main
3  {
4      public static void main(String[] args) {
5          int[] a = {2,2,1,2,3,2,2};
6          for(int i=0;i<a.length;i++){
7              int count=0;
8              for(int j=1;j<a.length;j++){
9                  if(a[i] == a[j]){
10                     count++;
11                 }
12                 System.out.print("Majority Element =" + a[i]);
13                 return;
14             }
15         }
16     }
17 }
18
19
20 }
21 }
22 }
```

input
Majority Element =2

...Program finished with exit code 0
Press ENTER to exit console.

17. Rearrange Array in Wave Form

Input:

1 2 3 4 5 6

Output:

2 1 4 3 6 5

```
Main.java
1 import java.util.*;
2 public class Main
3 {
4     public static void main(String[] args) {
5         int[] ar = {1,2,3,4,5,6};
6         int a=0,b=0;
7         for(int i=0;i<ar.length;i+=2){
8             a=ar[i];
9             ar[i]=ar[i+1];
10            ar[i+1]=a;
11        }
12        for(int i=0;i<ar.length;i++){
13            System.out.print(ar[i]+" ");
14        }
15    }
16 }
```

input

2 1 4 3 6 5

...Program finished with exit code 0
Press ENTER to exit console.

18. Find Longest Consecutive Increasing Sequence

Input:

1 2 3 1 2 3 4 5

Output:

Length = 5

```
Main.java
3
4 public class Main {
5     public static void main(String[] args) {
6
7         int[] a = {1,2,3,1,2,3,4,5};
8
9         int count = 1;
10        int max = 1;
11
12        for(int i = 0; i < a.length - 1; i++) {
13
14            int diff = a[i + 1] - a[i];
15
16            if(diff == 1) {
17                count++;
18            } else {
19                count = 1;
20            }
21
22            if(count > max) {
23                max = count;
24            }
25        }
26
27        System.out.println("Length = " + max);
28    }
29 }
```

input

Length = 5

...Program finished with exit code 0
Press ENTER to exit console.

19. Find Frequency of Every Element Without Using HashMap

Input:

1 2 2 3 1 4 2

Output:

1 -> 2

2 -> 3

3 -> 1

4 -> 1

```
Main.java
1 import java.util.*;
2
3 public class Main {
4     public static void main(String[] args) {
5
6         Scanner sc = new Scanner(System.in);
7
8         int n = sc.nextInt();
9         int arr[] = new int[n];
10        boolean visited[] = new boolean[n];
11
12        for(int i = 0; i < n; i++) {
13            arr[i] = sc.nextInt();
14        }
15
16        for(int i = 0; i < n; i++) {
17
18            if(visited[i]) {
19                continue;
20            }
21
22            int count = 1;
23
24            for(int j = i + 1; j < n; j++) {
25
26                if(arr[i] == arr[j]) {
27                    count++;
28                    visited[j] = true;
29                }
30            }
31
32            System.out.println(i + 1 + " -> " + count);
33        }
34    }
35}
```

7
1 2 2 3 1 4 2
1 -> 2
2 -> 3
3 -> 1
4 -> 1

20. Find the Smallest Positive Missing Number

Input:

3 4 -1 1

Output:

2

```
Main.java
1 import java.util.*;
2 public class Main {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         int n = sc.nextInt();
6         int arr[] = new int[n];
7         for(int i = 0; i < n; i++) {
8             arr[i] = sc.nextInt();
9         }
10        int ans = 1;
11        while(true) {
12            boolean found = false;
13            for(int i = 0; i < n; i++) {
14                if(arr[i] == ans) {
15                    found = true;
16                    break;
17                }
18            }
19            if(found) {
20                ans++;
21            } else {
22                break;
23            }
24        }
25        System.out.println(ans);
26    }
27 }
```

input

```
4
3 4 -1 1
2

...Program finished with exit code 0
Press ENTER to exit console.
```